



Department of Computer Science and Engineering
Final Examination Summer 2023
CSE 321: Operating Systems

Duration: 1 Hour 50 Minutes

Total Marks: 40

Answer the following questions.
 Figures in the right margin indicate marks.

1. CO4	a) Imagine a computer lab with multiple computers, and equipped with printers. Students use these computers for various tasks and may want to print documents. However, there are only a limited number of printers available. How do you solve this issue using semaphore? Your answer should have the steps associated in solving the given scenario.	[3]				
	b) What is meant by busy waiting in the implementation of a critical section solution, and why is it considered problematic?	[1.5]				
	c) For Peterson's problem below conditions will be applied. • There are two processes: P1 and P2. P2 gets to execute first. • Each Statement takes 4ms to execute. • Context Switch will occur after 12ms. • Critical section contains 4 statements. • Remainder section contains 3 statements. • For P1: i=0 and j=1 • For P2: i=1 and j=0 • turn=0 • flag[0] = FALSE, flag[1] = TRUE The structure of process P_i in Peterson's solution: <pre>do{ flag[i] = true; turn = j; while(flag[j] == true && turn == 1){ //busy wait } //critical section flag[i] = false; //remainder section }while(true);</pre>					
	Complete the table below for processes P₁ and P₂ using Peterson's solution. <table><tr><td>Process 1: i = 0, j = 1</td><td>Process 2: i = 1, j = 0</td></tr><tr><td></td><td></td></tr></table>	Process 1: i = 0, j = 1	Process 2: i = 1, j = 0			[3.5]
Process 1: i = 0, j = 1	Process 2: i = 1, j = 0					

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2. CO4	a) Consider the following snapshot of a system: <table><tr><td></td><td></td><td colspan="4">Max</td><td></td><td colspan="4">Allocation</td><td></td><td></td><td colspan="4">Available</td></tr><tr><td></td><td></td><td>A</td><td>B</td><td>C</td><td>D</td><td></td><td>A</td><td>B</td><td>C</td><td>D</td><td></td><td></td><td>A</td><td>B</td><td>C</td><td>D</td></tr><tr><td>P1</td><td></td><td>7</td><td>0</td><td>1</td><td>3</td><td></td><td>7</td><td>0</td><td>0</td><td>2</td><td></td><td></td><td>1</td><td>6</td><td>4</td><td>4</td></tr><tr><td>P2</td><td></td><td>2</td><td>7</td><td>5</td><td>0</td><td></td><td>2</td><td>1</td><td>0</td><td>0</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>P3</td><td></td><td>2</td><td>12</td><td>5</td><td>6</td><td></td><td>0</td><td>6</td><td>3</td><td>3</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>P4</td><td></td><td>1</td><td>6</td><td>5</td><td>6</td><td></td><td>0</td><td>2</td><td>1</td><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>			Max					Allocation						Available						A	B	C	D		A	B	C	D			A	B	C	D	P1		7	0	1	3		7	0	0	2			1	6	4	4	P2		2	7	5	0		2	1	0	0							P3		2	12	5	6		0	6	3	3							P4		1	6	5	6		0	2	1	2							
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P4		1	6	5	6		0	2	1	2																																																																																														
	i. Calculate the Need Matrix. Is this system in a safe state? If yes, then find the safe sequence using Banker's Safety algorithm otherwise, provide the necessary explanation.	[1+3]																																																																																																						
	ii. What happens if the process P4 requests at this moment for (0, 4, 2, 0)? Whether Banker's algorithm grants the request or not?	[5]																																																																																																						
	b) Suppose, in a workplace, we have a set of resource types, $R = \{R1, R2, R3, R4\}$ and a set of processes, $P = \{P1, P2, P3, P4, P5\}$. R1, R2, R3, and R4 have 3, 2, 3, and 2 instances respectively. • P1 is holding 2 instances of R1 • P2 is holding 1 instance of R3 • P3 is holding 1 instance of R4 • P5 requests 2 instances of R3 • P4 is holding 1 instance of R4 • P3 requests 1 instance of R2 • P2 requests 1 instance of R1 • P2 is holding 1 instance of R2 • P1 is requesting 1 instance of R4 • P3 is holding 1 instance of R3 • P4 is holding 1 instance of R2																																																																																																							
	Construct a resource allocation graph for the above scenario and identify the cycle (if any) and decide whether there is a deadlock or not.	[4]																																																																																																						

3. CO5	a) Given variable size memory (dynamic) partitions of 8 MB, 29 MB, 35 MB, and 48 MB (in order, top to bottom), apply best-fit and worst-fit algorithms to place processes with the space requirement of 5 MB, 15 MB, 10 MB, 5 MB, 10 MB, 20 MB, 25 MB, and 15 MB (in order). Which algorithm makes the most effective use of memory?	[4+1]
	b) How is paging efficiently used in main memory to increase throughput? Justify with necessary examples.	[3]
	c) Compare the following systems in terms of performance: 1. A system with a hit ratio of 93%, associative lookup time of 32ns, and memory access time of 72ns. 2. A system with a hit ratio of 62%, associative lookup time of 9ns, and memory access time of 133ns.	[3]
4. CO5	a) Suppose there are two processes P1 (32 B) and P2 (40 B) with a page size of 8 B . The main memory size is 96 B . The page table for P1 and P2 are given below: Page table: P1 Page table: P2	
	Find the corresponding physical addresses of the following logical addresses: a) Address 011010 of P1 b) Address 111110 of P2 c) Address 011001 of P2	[3]
	b) Consider a computer with a main memory that has 5 frames and page reference string of 0-7 page [2, 2, 2, 6, 5, 5, 4, 2, 0, 0, 6, 1, 5, 5, 3, 0, 0, 2, 6, 5] . The page reference string represents the order in which the pages are accessed by a program. Apply LRU & OPT algorithm to simulate the page replacement that occurs when the main memory can hold at most 5 pages at a time. Record the number of page faults and compare the result. Mention which algorithm performs better in this scenario.	[4+1]

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